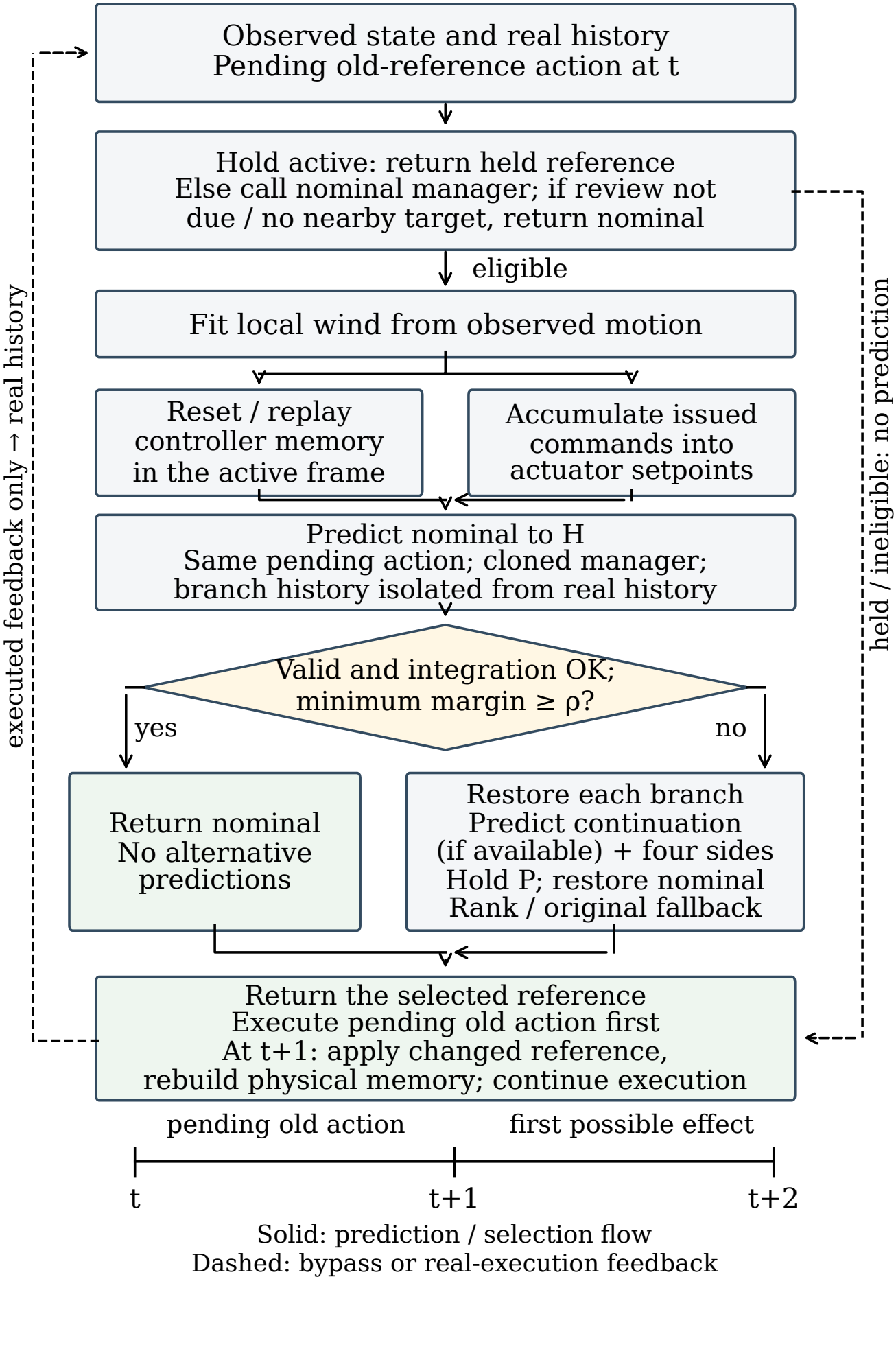
Observed state and real history
Pending old-reference action at t
Hold active: return held reference
Else call nominal manager; if review not due / no nearby target, return nominal
eligible
Fit local wind from observed motion
Reset / replay controller memory in the active frame
Accumulate issued commands into actuator setpoints
Predict nominal to H
Same pending action; cloned manager; branch history isolated from real history
Valid and integration OK; minimum margin ≥ ρ?
yes
no
Return nominal
No alternative predictions
Restore each branch
Predict continuation (if available) + four sides
Hold P; restore nominal
Rank / original fallback
Return the selected reference
Execute pending old action first
At t+1: apply changed reference, rebuild physical memory; continue execution
executed feedback only → real history
held / ineligible: no prediction
pending old action
first possible effect
t
t+1
t+2
Solid: prediction / selection flow
Dashed: bypass or real-execution feedback